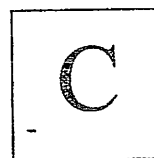


--	--	--	--	--	--	--	--



B.Tech. Degree I Semester Supplementary Examination June 2024

CE/ME/SE 23-200-0102 A ENGINEERING CHEMISTRY (2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Explain the basic concepts of chemical thermodynamics and quantum chemistry.
 CO2: Illustrate the spectroscopic methods in characterizing materials.
 CO3: Develop electrochemical methods to protect different metals from corrosion.
 CO4: Interpret the chemistry of a few important engineering materials and their industrial applications.
 CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

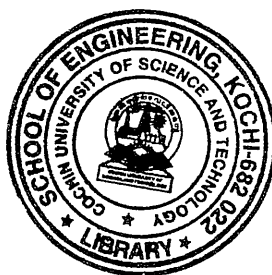
PART A (Answer ALL questions)

	(2 × 5 = 10)	Marks	BL	CO	PO
I. (a) Explain the first law of thermodynamics with mathematical expression.	2	2	L1	1	3
(b) Write down all the quantum numbers for (i) valence electron of Potassium (Z = 19) (ii) valence orbit of Neon (Z=10).	2	2	L2	1	2
(c) Write the principle behind microwave spectroscopy.	2	2	L1	2	1
(d) Explain the significance of Nernst equation.	2	2	L1	3	1
(e) Write a note on the salient features of refractories.	2	2	L1	4	3

PART B

(4 × 10 = 40)

II.	Explain Pattinson's process with a neat labelled phase diagram.	10	L2	1	2
OR					
III. (a)	Derive the relation $\Delta G^\circ = RT \ln K_p$.	7	L2	5	3
(b)	Obtain the equilibrium constant K_{eq} for the following reaction at 400 K. $2\text{NOCl} \rightleftharpoons 2\text{NO(g)} + \text{Cl}_2\text{(g)}$ Given that $\Delta H^\circ = 77.2 \text{ KJ mol}^{-1}$ and $\Delta S^\circ = 122 \text{ JK}^{-1} \text{ mol}^{-1}$.	3	L2	1	2
IV.	Derive the operator form of Schrodinger equation from classical wave equation.	10	L1	1	3
OR					
V. (a)	Obtain the forms of hydrogen atom wave functions.	6	L2	1	3
(b)	For the wavefunction $\psi = A \cos x$ for $0 < x < \pi/2$. Find the value of A.	4	L3	1	3



(P.T.O)

BTS-I(S)-06-24-2496

		Marks	BL	CO	PO
VI.	(a) Discuss the possible electronic transitions, their regions of absorption and shifts in the position of absorption in electronic spectroscopy.	7	L2	2	1
	(b) TMS is used as internal reference in NMR. Why?	3	L3	2	3
OR					
VII.	(a) Draw the low and high resolution ^1H NMR spectra of	6	L3	3	1
	(i) $\text{CH}_3\text{-CH}_2\text{-CH-Cl}_2$				
	(ii) $\text{CH}_3\text{-CCl}_2\text{-CH}_3$				
	(iii) $\text{C}_6\text{H}_5\text{-CH}_2\text{-CH}_3$				
	(b) The fundamental vibrational frequency of HI^{127} molecule is $6.6 \times 10^{13} \text{ s}^{-1}$ and the internuclear distance is $1.5 \times 10^{-8} \text{ m}$. Calculate the relative force constant.	4	L3	2	3
VIII.	(a) Write a note on the effects of corrosion and methods to prevent corrosion.	5	L2	3	1
	(b) Explain the different physical properties to benchmark Lubricants.	5	L2	4	3
OR					
IX.	(a) Describe the mechanism of lead storage battery.	5	L2	3	1
	(b) Write a note on conducting polymers.	5	L2	4	3

Bloom's Taxonomy Levels

L1-22.7%, L2-59%, L3-18.1%.

